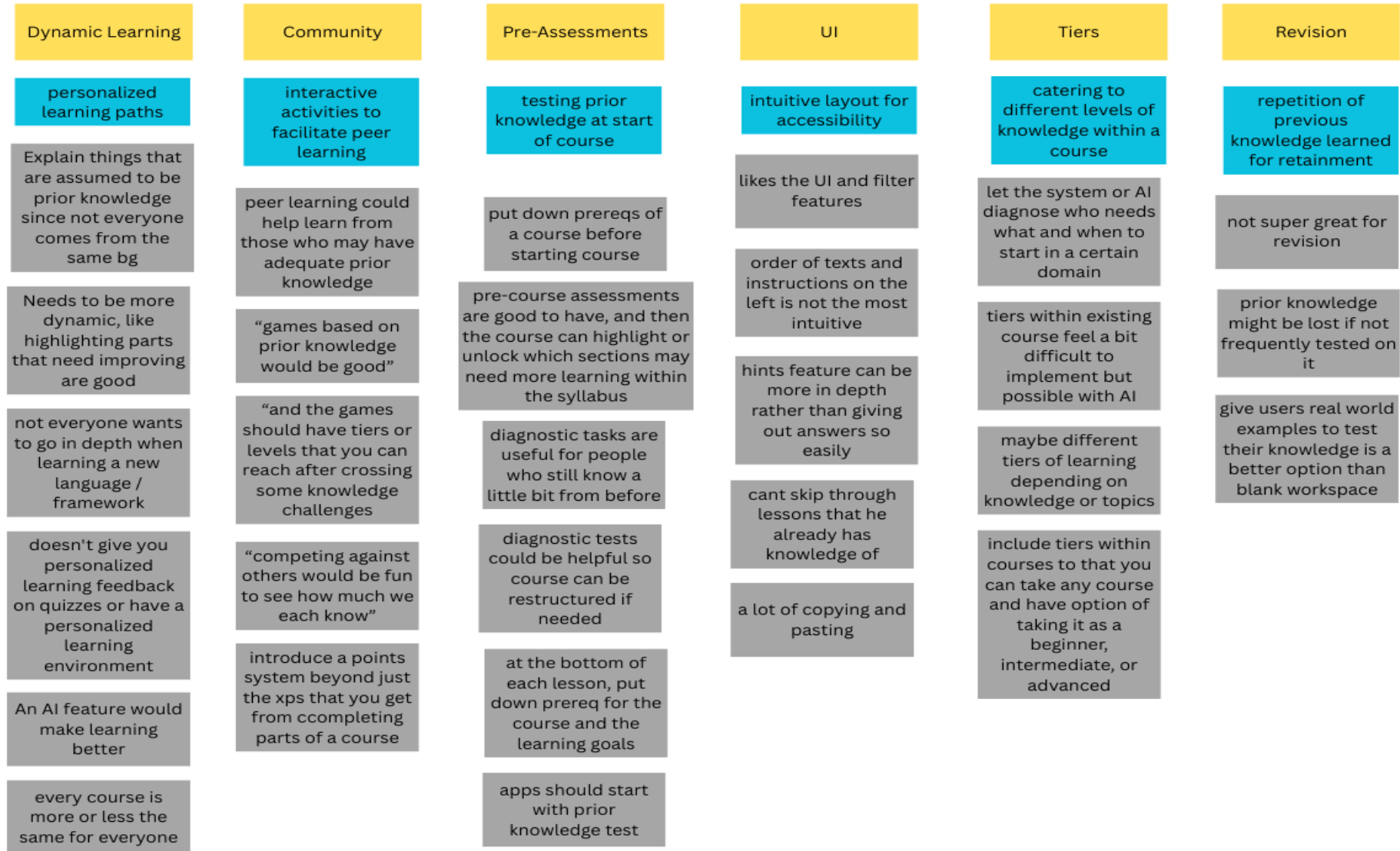


Prior Knowledge Group 24

Ishika Patel, Joyce Li, Jiajie Qian, Hamzah Zahid

Affinity Diagram – Khan Academy

Khan Academy



Problems with Khan Academy

1. No option to skip or adjust content based on existing understanding
2. No diagnostic or pre-course assessment to help place students at the right level
3. Limited community features for students to learn collaboratively
4. No built-in system for spaced repetition and/or revision within the course flow
5. No flexibility to adjust the difficulty tier (beginner/intermediate/advanced) dynamically
6. No fun way to interact with course learning goals

Original Design Ideas

1. Learning Tiers within Courses

- Adaptive Learning Path – Choose from Beginner, Intermediate, or Advanced
- Users select known topics to personalize course content

2. Pre-Course Assessments

- Allows users who have prior knowledge to get tested to be placed into the proper course path
- Perfect for users who don't know how much they know

3. Peer Learning Community

- Gamified approach allowing users with prior knowledge to reinforce and accurately calibrate against global users

4. Smart Revision Tools

- Allows users to retain knowledge across lessons, and apply it across a range of concepts

Feature 1

Course Tiers

Background Knowledge

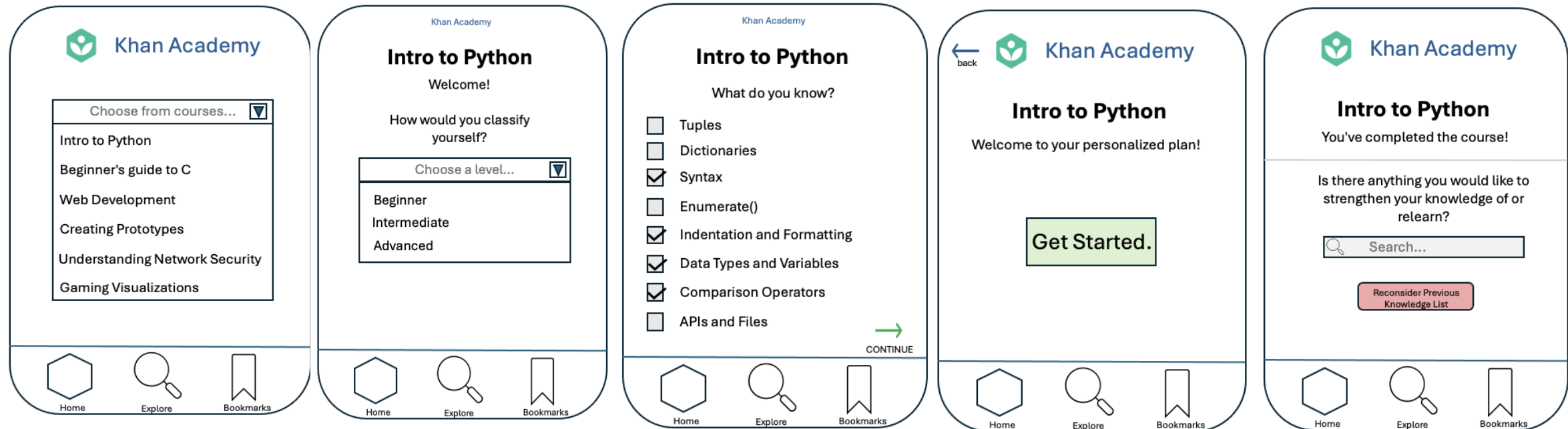
- ***How Learning Works* – Chapter 1**

- Students' prior knowledge can help or hinder learning, depending on whether it's accurate, sufficient, or relevant.
- **What we are doing:**
 - Allowing users to self-assess and select their own tier and provide a way for users to identify which topics they already know.
 - Allowing users to retake the course at the end in case they provided an inaccurate list of topics that they already know.

- **Learning Principles:**

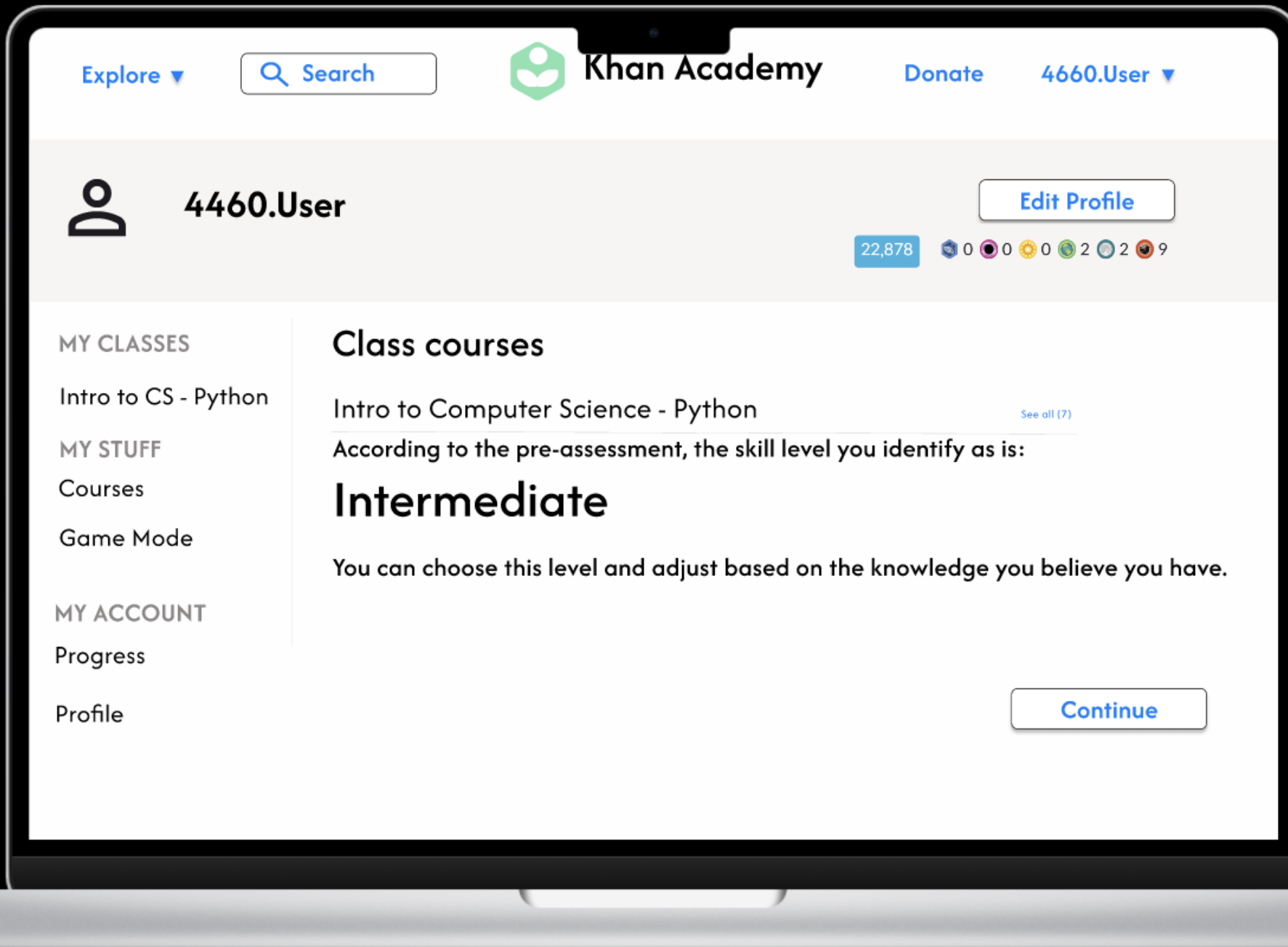
- Learning is culturally embedded, and prior experiences vary among learners
- Learners actively build knowledge based on what they already know and their context

Testing Phases: Feature 1 – Learning Tiers



User Testing Feedback:

1. Make the search bar more contextual
2. Change wording on the last screen so that it is more apparent to the user of what the button does



TA Feedback:
"It would be nice to see the computer providing recommendations for skill level based on the user's pre-assessment, because users may not often know where to start."



4460.User

Edit Profile

22,878

0 0 0 2 2 9

MY CLASSES

Intro to CS - Python

MY STUFF

Courses

Game Mode

MY ACCOUNT

Progress

Profile

Class courses

Intro to Computer Science - Python

[See all \(7\)](#)

What do you already know?

Based on the category you choose (beginner, intermediate, or advanced), Khan Academy will readjust the course for you.

1. Choose your desired skill level.
2. Check the concepts you believe you do not need in your learning journey.
3. Let the readjustments align with your goals. If you are not satisfied, please
· refine the skill level or the concepts you have chosen to skill.

Fill out the information below to get your personalized learning plan.

Skill Level ▲

Beginner

Intermediate

Advanced

[Courses](#)[Game Mode](#)

MY ACCOUNT

[Progress](#)[Profile](#)

What do you already know?

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Fill out the information below to get your personalized learning plan.

Intermediate ▲

Beginner

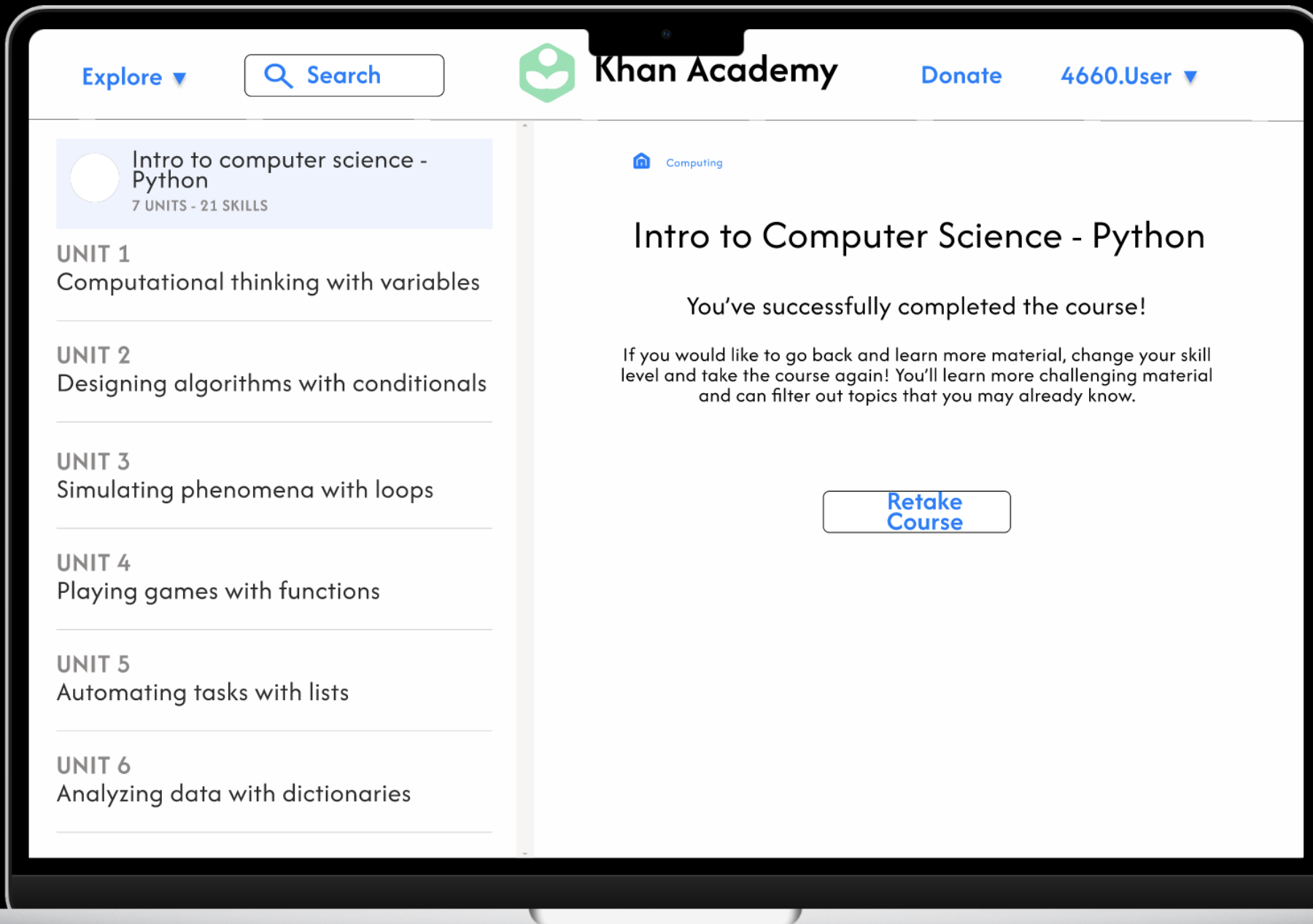
Intermediate

Advanced

What knowledge do you already have?

- ☒ Lambda Functions
- ☒ Classes and Object-Oriented Programming
- ☐ File Handling
- ☐ Sorting and Searching Algorithms
- ☐ Recursion

[Go Back](#)[Continue](#)



Improvement:

- No more search bar
- Made button name more concise
- Simplified for users
 - o Now they don't to overthink all of the possible topics themselves

MY CLASSES

Intro to CS - Python

MY STUFF

Courses

Game Mode

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Progress

Profile

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[See all \(7\)](#)

What do you already know?

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· refine the skill level or the concepts you have chosen to skill.

Fill out the information below to get your personalized learning plan.

Beginner ▲

Beginner

Intermediate

Advanced

What knowledge do you already have?

- ☐ variables and data types
- ☐ conditional statements
- ☐ operators
- ☐ string manipulation
- ☐ input() and print()

Feature 2

Pre-Assessments

Background Knowledge

- ***How Learning Works* – Chapter 1**

- Understanding what students know/think they know can help instructors design their instruction appropriately in a way that identifies and fills gaps and works towards correcting misconceptions
- Great way to test prior knowledge is to administer a diagnostic assessment
- Users are placed in the best portion of the course based on the assessment

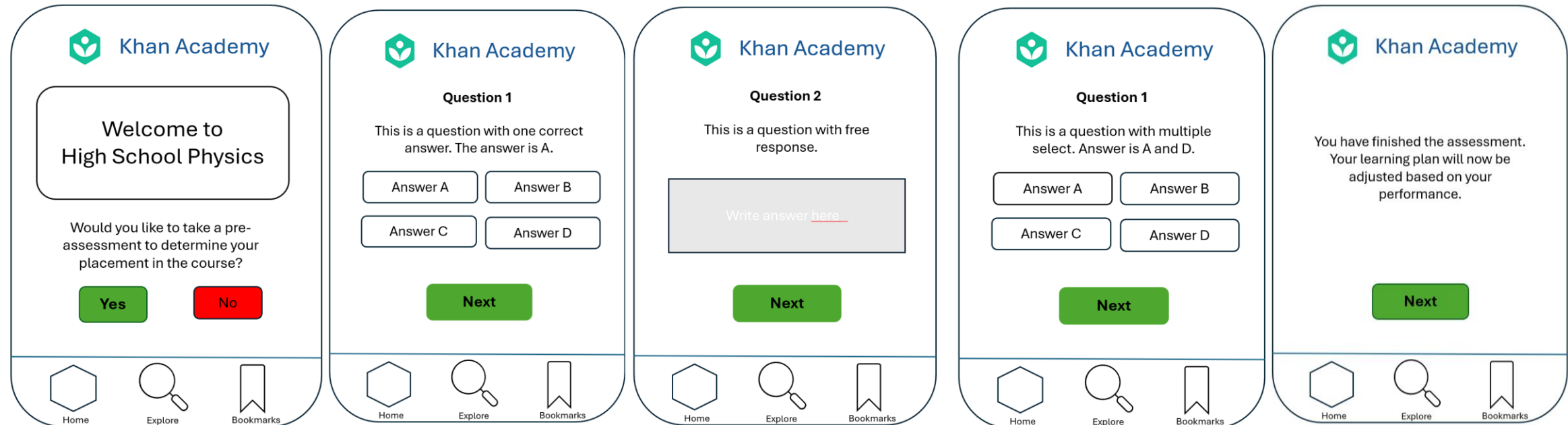
- **Hailikari et al. (2006)**

- Many instructors found a positive correlation between performance in prior knowledge test and the final grade
- Students who know and understand their prior knowledge will do better on the course than students who do not

- **Learning Principles:**

- Creating different types of questions that test similar topics in different manners tests the procedural and metacognitive knowledge, ensuring students fully understand the material
- Using diagnostic assessments allows users to remember the knowledge they previously learned and apply said knowledge, allowing for easier retrieval later in the course

Testing Phases: Feature 2 – Pre-Assessments



User Testing Feedback:

1. The multiple select is confusing and should be formatted differently from a multiple choice.
2. There should be an assessment results page that tells you how you did and what questions you got wrong.

Question 1

What type of data is this: 4.01 (US), 4,01(EU)?

float

boolean

integer

string

Question 2

What is the difference between a double and a float?

double is more precise
while float is less precise

Question 3

When following the rules of encapsulation, what methods do you have to create to access and modify data?

- ☒ Getters ☐ Child
☐ Parent ☐ Setters

Question 4

This is a fill in the blank question.

_____ is the idea of bundling data and methods into easy-to-use units.

encapsulation

Question 5

Please group similar concepts into the same category

Concept A

Concept C

Concept B

Concept D

Here are the results from the assessment:

Topic A: 4/4 Questions Correct

Topic B: 4/4 Questions Correct

Topic C: 2/5 Questions Correct

Based on the following, we have placed in you in Unit 4.

Improvements:

- Added in actual questions to simulate a pre-assessment
- Added in two more question types: fill in the blank and grouping
- Added a results screen along with different colors to show how they did on each topic

Feature 3

Game/Community Learning

Background Knowledge

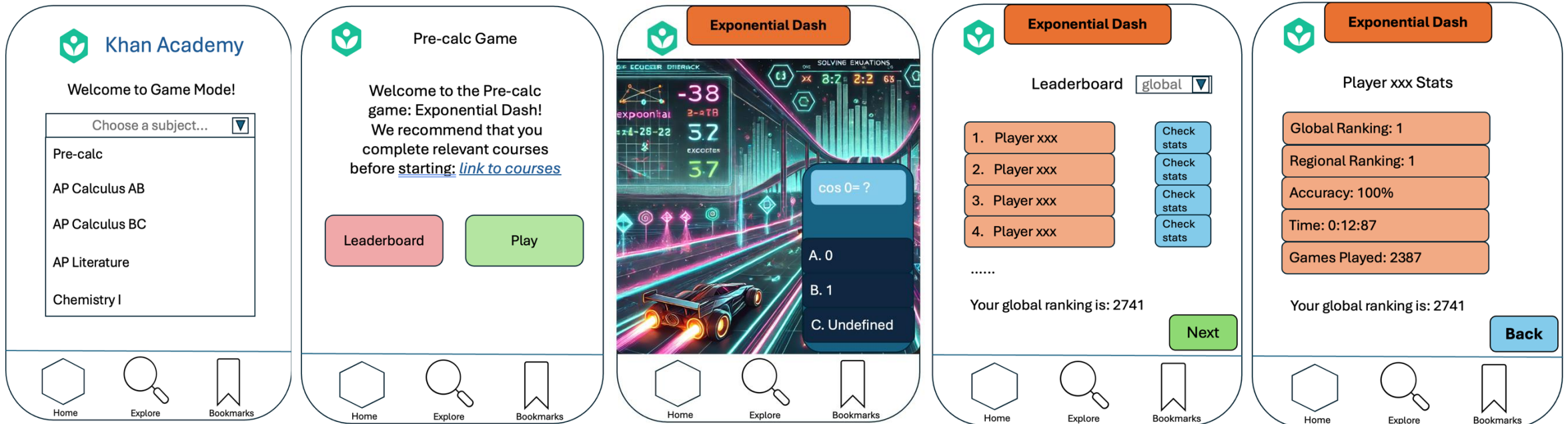
- **Hailikari et al. (2006)**

- Learning is influenced by the interplay of cultural, social, cognitive, and biological contexts
- Encourages users to demonstrate their skills through gaming challenges
- The game promotes social interactions through community features where players gain strategies and insights on what to focus on (speed, accuracy, games played etc.), helping to reveal their level of competence.
- Allows users to replay and improve their overall accuracy

- **Learning Principles:**

- Players set goals, track progress, and reflect on their achievements, developing metacognitive skills crucial for independent learning
- Learners build understanding through meaningful interactions and hands-on engagement rather than passively absorbing information
- The game environment is designed to encourage learners to apply concepts in new contexts, reinforcing the idea that understanding goes beyond memorization

Testing Phases: Feature 3 – Game/Community Learning



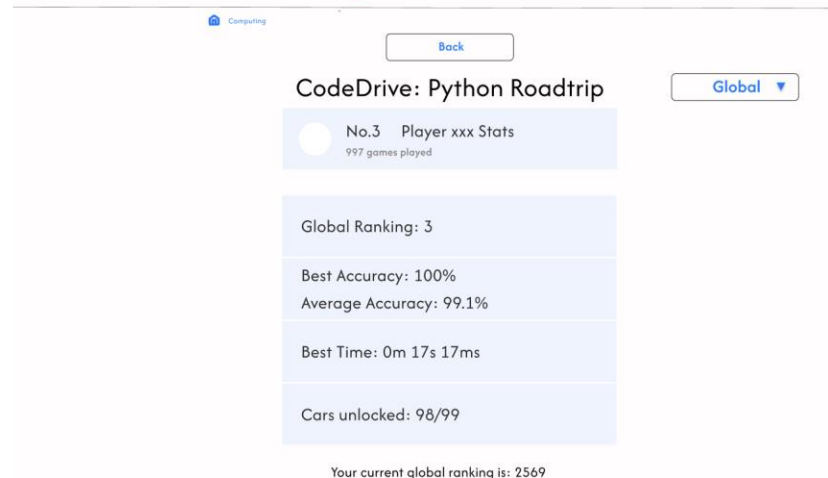
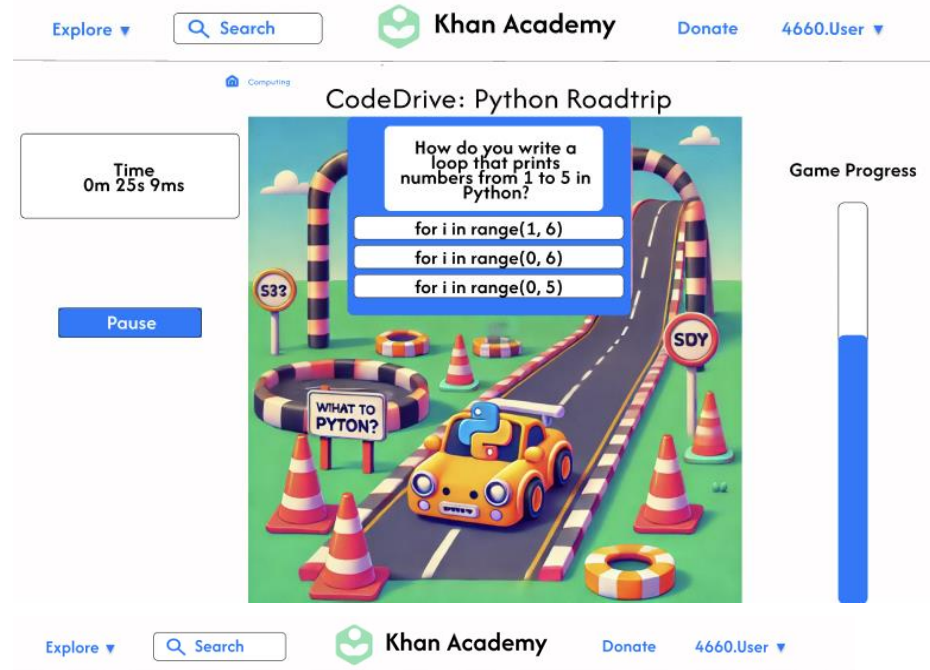
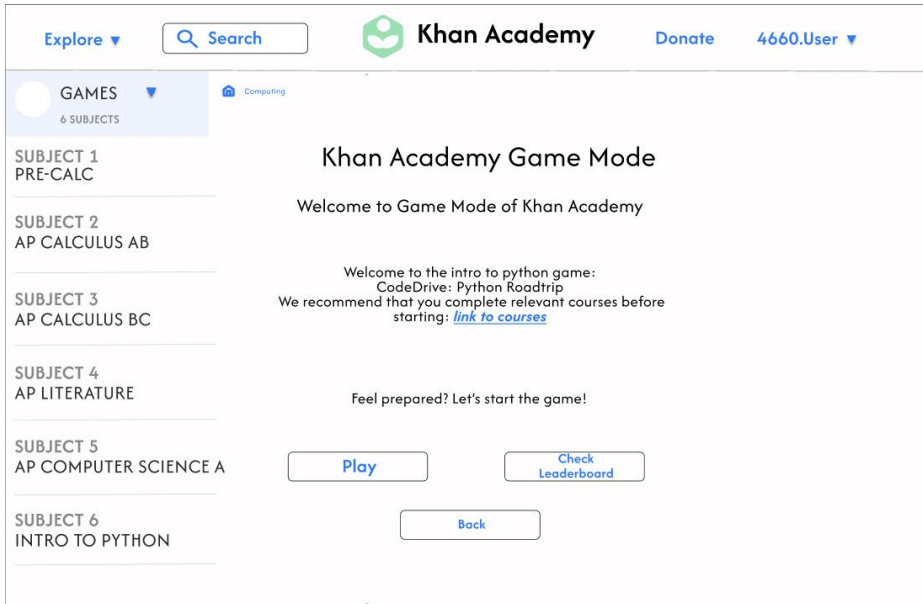
User Testing Feedback:

1. A level-up mechanism with rewards such as unlocking a new car to enhance further engagement.
2. More intuitive leaderboard allowing users to view stats of multiple players to identify areas of focus

Feature 3 Demo – Prototype 1

Improvement:

- Links to necessary learning material
- Progress bar during game
- More information on how ranking is calculated, hinting users areas of focus
- o All align with goal-directed practice learning science principle



Feature 4

Revision

Background Knowledge

- *How People Learn*

- Active prior knowledge of a topic can help students integrate new information.
- Allows users to activate and retain their prior knowledge while linking it to newer concepts.

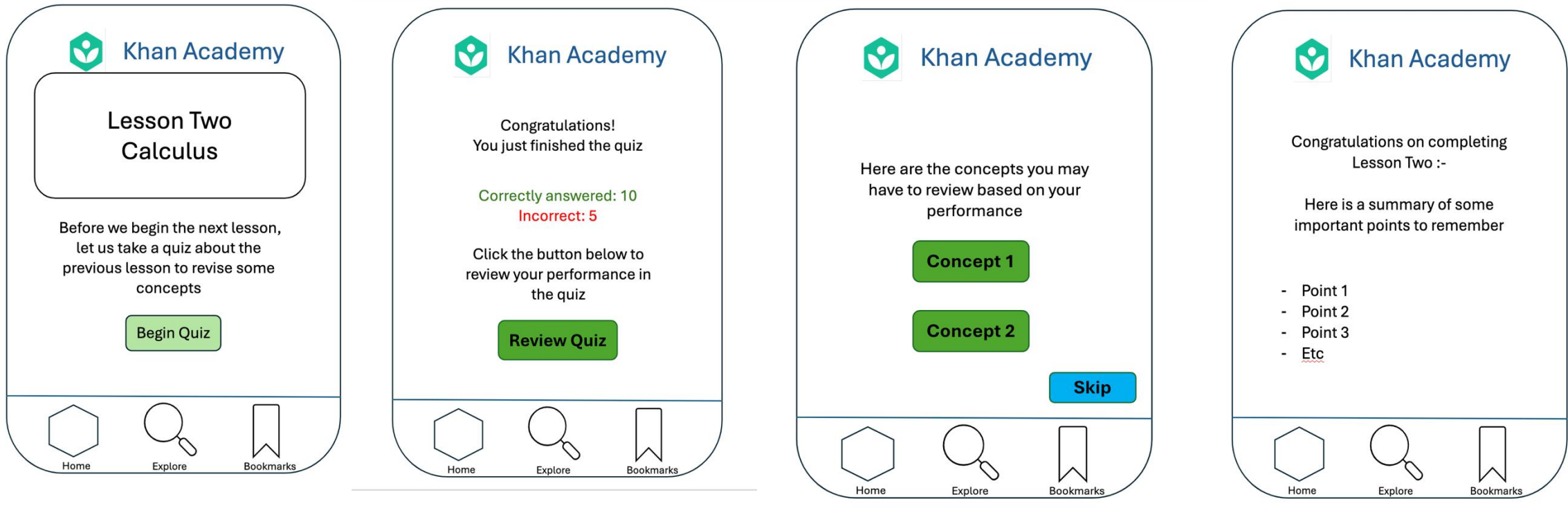
- **Hailikari et al. (2006)**

- It is also important to give feedback on assessments, as students will be able to learn what they got wrong and do better for the actual assessment.
- The revision quiz focuses on analyzing the weaker topics and allowing users to revise those concepts more deeply

- **Learning Principles:**

- Learners are reminded and tested on the skills learnt in the previous lessons, and then connecting those ideas to new knowledge, allowing interleaving to take place
- Learners are motivated through constructive feedback, as well as an autonomy-based revision process which allows for stronger motivation

Testing Phase : Feature 4 - Revision



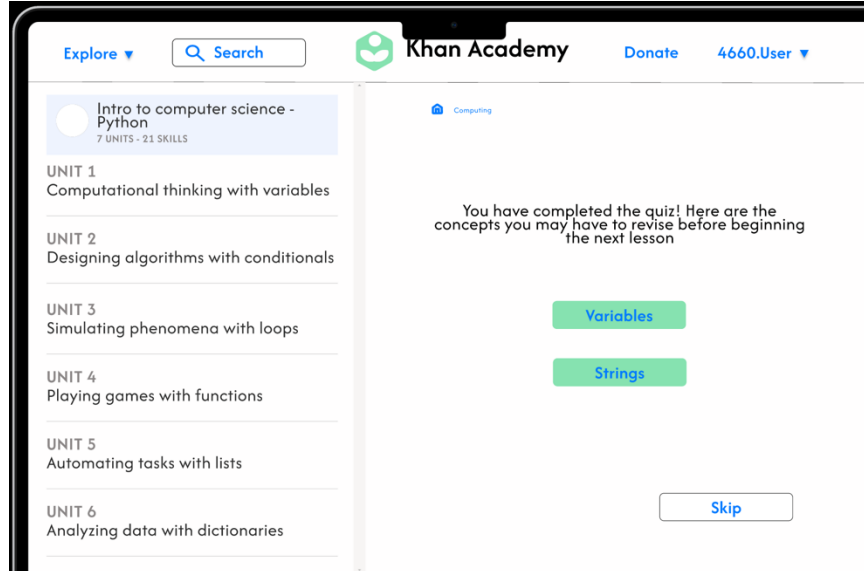
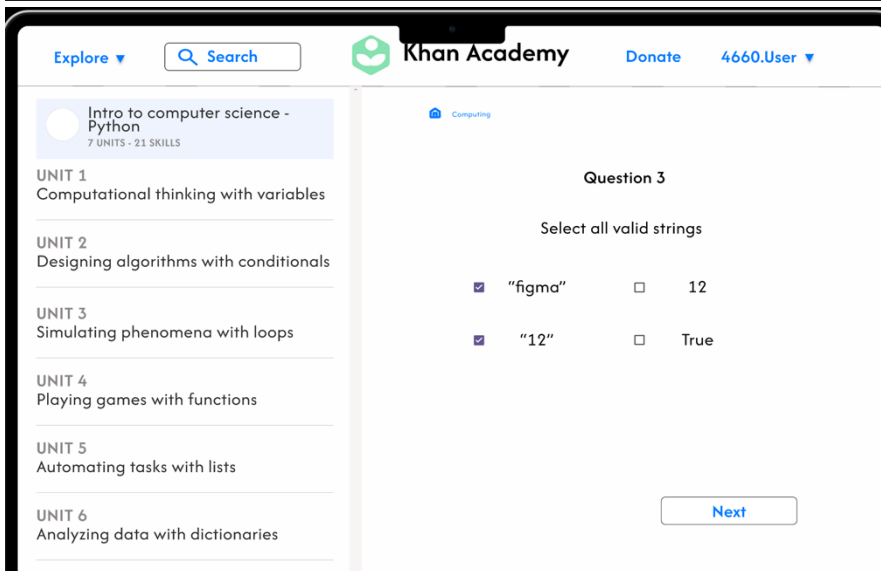
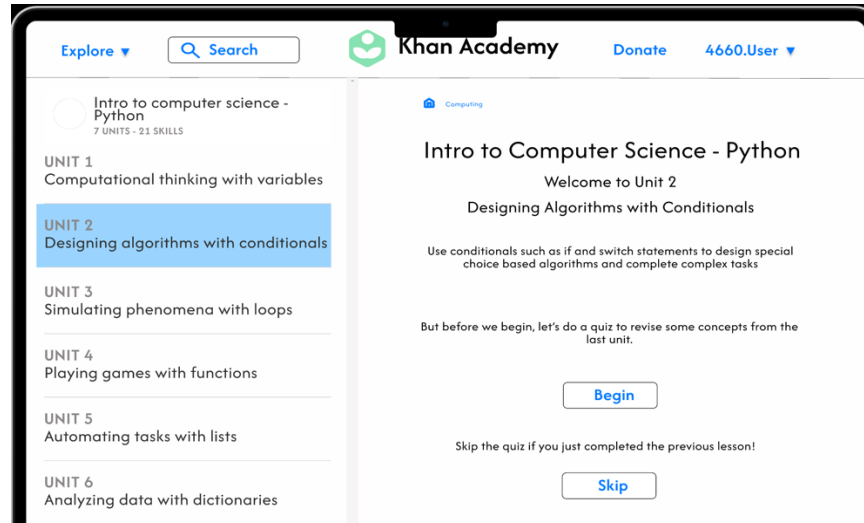
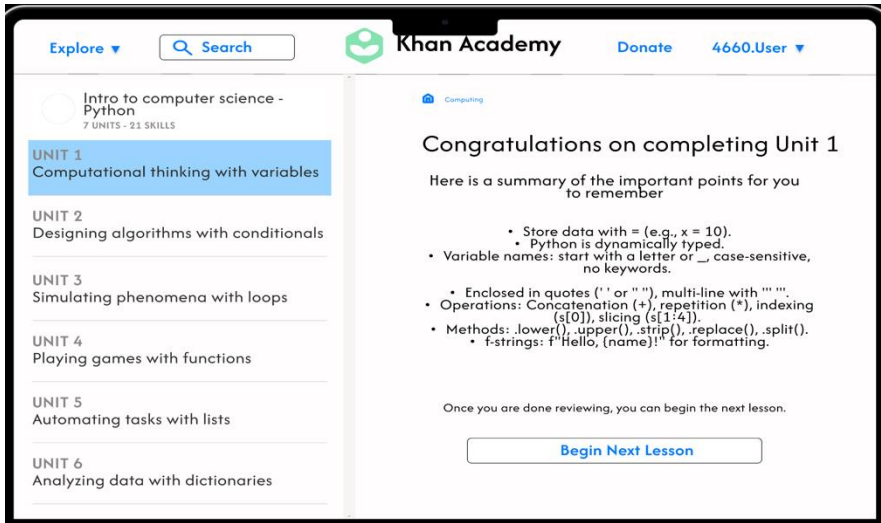
User testing feedback:

- Detailed feedback on incorrect answers from the quiz would be helpful
- Include scenario-based questions at the end for stronger retention.

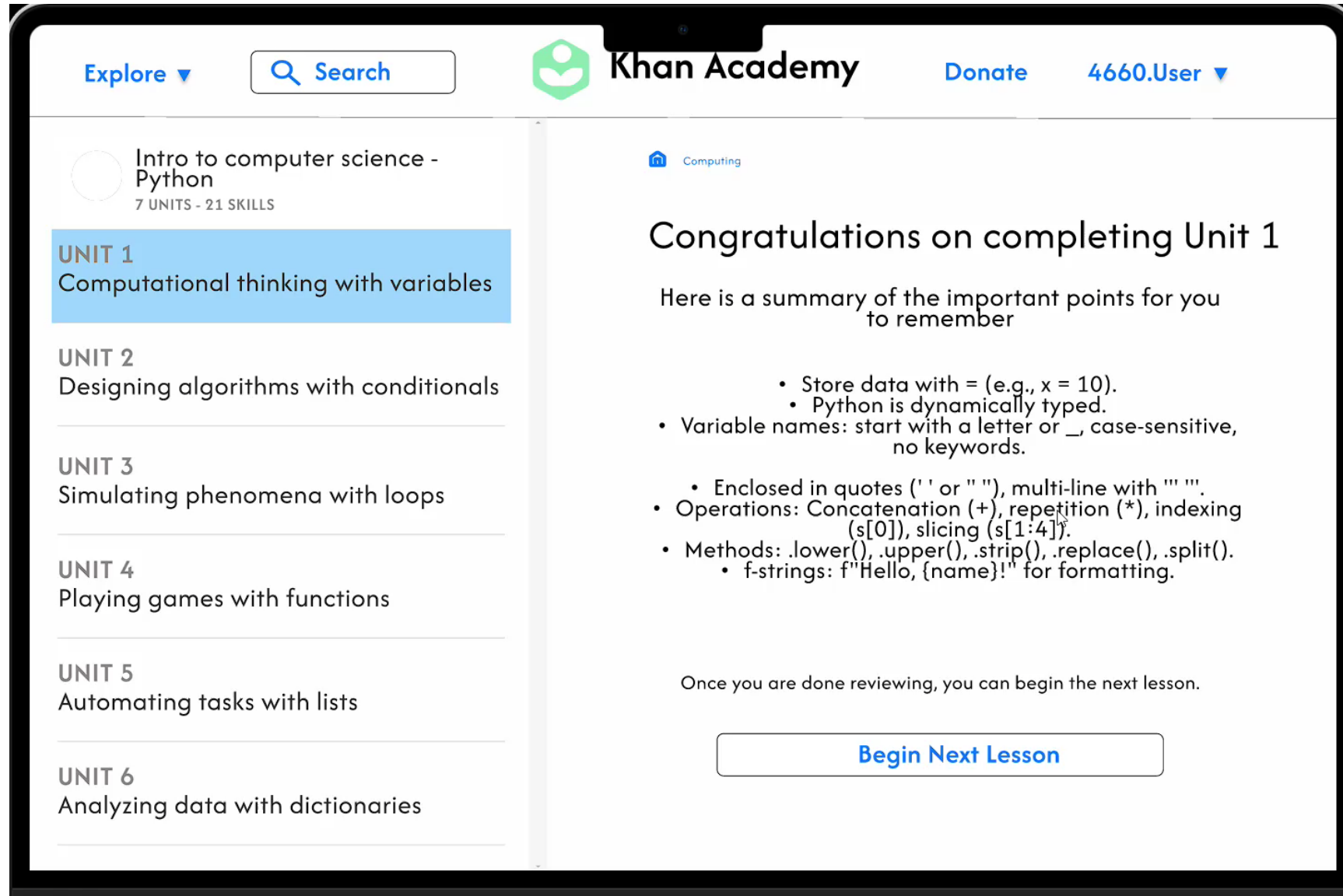
Feature 4 Demo – Prototype 1

Improvements:

- Provided detailed summary at the end of each lesson
- Ability to skip the quiz if just completed lesson
- Improved quality of questions to connect and transition more smoothly to the new lesson



Full Video Demo and Figma Link



[Figma link](#)

Thank You

Group 24